

PHISHING EMAIL DETECTION & AWARENESS REPORT

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FutureInterns Cybersecurity Internship Task 2

1. Introduction

Phishing is one of the most common and costly cyber attacks faced by individuals and businesses today. Attackers craft deceptive emails to trick users into clicking malicious links, downloading infected files, or surrendering sensitive credentials such as passwords and OTPs.

According to industry research, phishing is responsible for over 90% of data breaches worldwide. Despite advances in technical security controls, human awareness remains the most critical line of defense. This report analyzes three real-world phishing email samples, identifies their attack indicators, and provides practical guidance that employees can immediately apply.

This report is intended for educational and security awareness purposes only. No systems were exploited or compromised during this analysis.

2. Methodology

Samples Analyzed	3 phishing email samples
Tools Used	Google Apps Header Analyzer, MXToolbox Email Header Analyzer, Manual URL inspection, Browser DevTools
Analysis Criteria	Sender domain verification, Header authentication (SPF/DKIM/DMARC), URL and link inspection, Language and content pattern analysis, Attachment identification
Classification Scale	PHISHING = 3+ confirmed indicators SUSPICIOUS = 1-2 minor flags SAFE = No indicators found

3. Findings Overview

All three email samples were confirmed as phishing attempts. The table below summarizes the key indicators identified across each sample.

#	Finding	Indicator Type	Risk
1	Fake sender domain impersonating a bank	Domain Spoofing	HIGH
2	Reply-To redirects to attacker-controlled address	Header Manipulation	HIGH
3	Urgency / fear-based subject line	Social Engineering	HIGH
4	Generic greeting — no real name used	Personalization Failure	MEDIUM
5	Malicious link using HTTP and fake domain	Malicious URL	HIGH
6	SPF / DKIM / DMARC authentication failure	Header Auth Failure	HIGH
7	Fake IT helpdesk impersonation	Domain Spoofing	HIGH
8	Password expiry threat to force action	Social Engineering	HIGH
9	Link redirects to credential harvesting portal	Malicious URL	HIGH
10	Fake DHL delivery notification	Brand Impersonation	HIGH
11	Small payment request to lower victim suspicion	Financial Lure	MEDIUM
12	Reply-To points to .ru domain	Header Manipulation	HIGH

HIGH RISK 10 indicators	MEDIUM RISK 2 indicators	LOW RISK 0 indicators
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4. Sample Analysis

Sample 1 — Fake Bank Security Alert

CLASSIFICATION

HIGH

From: security@bankofamerica-alerts.net
Reply-To: collect@mail-support.ru
Subject: Urgent: Your Account Has Been Suspended

Dear Customer,

We have detected unusual login activity on your Bank of America account. Your account has been temporarily suspended for your protection.

To restore access, verify your identity immediately:
>> Click Here: <http://boa-secure-login.net/verify>

You have 24 hours. Failure to comply will result in permanent account closure.

Regards,
Bank of America Security Team

Indicators Found

Fake Sender Domain	bankofamerica-alerts.net is NOT bankofamerica.com — common spoofing tactic
Reply-To Mismatch	Replies go to collect@mail-support.ru — a completely different attacker domain
Urgency Language	Subject: 'Urgent' + 'Suspended' — designed to cause panic and rushed action
Generic Greeting	'Dear Customer' — real banks address you by full name
Malicious Link	http://boa-secure-login.net — uses HTTP (not HTTPS), fake domain
Header Auth	SPF: FAIL DKIM: FAIL DMARC: FAIL

How This Attack Works

The attacker impersonates Bank of America's security team. The victim panics about their account being suspended and clicks the link without thinking. They are taken to a convincing fake login page where they enter their real bank credentials — which are immediately captured by the attacker.

Sample 2 — Fake IT Password Reset

CLASSIFICATION

HIGH

From: it-helpdesk@yourcompany-support.net
Reply-To: reset@mail-helpdesk.xyz
Subject: ACTION REQUIRED: Your Password Expires Today

Dear Employee,

Your company email password will expire in 2 hours.

To avoid losing access to your account, reset your password immediately using the link below:

>> Reset Password Now: <http://reset-portal-secure.xyz/login>

If you do not reset, your account will be locked and you will need to contact IT manually.

This is an automated message. Do not reply to this email.

IT Support Team
yourcompany-support.net

Indicators Found

Fake Sender Domain	yourcompany-support.net is NOT a legitimate corporate IT domain
Reply-To Mismatch	reset@mail-helpdesk.xyz — unrelated domain, attacker-controlled
Urgency Language	'ACTION REQUIRED' + 'Expires Today' + '2 hours' — extreme time pressure
Malicious Link	http://reset-portal-secure.xyz — HTTP only, fake domain designed to look official
Discourages Contact	'Do not reply' — attempts to prevent victim from verifying with real IT team
Header Auth	SPF: FAIL DKIM: FAIL DMARC: FAIL

How This Attack Works

This attack targets employees in a workplace environment. The fake IT email creates extreme urgency around password expiration. The victim follows the link and enters their work credentials on a fake portal. The attacker gains access to the employee's corporate account, potentially exposing internal systems and data.

Sample 3 — Fake Package Delivery (DHL)

CLASSIFICATION

HIGH

From: delivery@dhl-trackingportal.com
Reply-To: noreply@dhl-parcels.ru
Subject: Your Package Could Not Be Delivered — Action Required

Dear Customer,

We attempted to deliver your package today but were unable to complete the delivery.

To reschedule, please confirm your address and pay a small redelivery fee of \$1.99:

>> Reschedule Delivery: <http://dhl-redelivery-portal.com/track>

Your package will be returned to sender within 48 hours if no action is taken.

Tracking Number: DHL-48271930

DHL Customer Service

Indicators Found

Fake Sender Domain	dhl-trackingportal.com is NOT dhl.com — brand impersonation
Reply-To Mismatch	noreply@dhl-parcels.ru — .ru domain, clearly unrelated to DHL
Financial Lure	\$1.99 fee — small amount used to lower victim suspicion and collect card details
Generic Greeting	'Dear Customer' — DHL uses your registered name
Fake Tracking Number	DHL-48271930 looks real but is fabricated to add legitimacy
Malicious Link	http://dhl-redelivery-portal.com — HTTP, fake domain to steal payment details
Header Auth	SPF: FAIL DKIM: FAIL DMARC: FAIL

How This Attack Works

This is a smishing-style email attack using a well-known brand. The victim expects a real delivery and is tricked by the realistic tracking number. The small \$1.99 fee reduces suspicion. When the victim enters their card details to pay the fee, the attacker captures their full payment information.

5. Common Phishing Indicators

The following indicators were observed across all three samples and represent the most common tactics used in phishing attacks:

Indicator	What It Looks Like	Risk Level
Fake sender domain	paypal-secure.com instead of paypal.com	HIGH
Urgency / fear language	'Act within 24 hours or lose access'	HIGH
Generic greeting	'Dear User' instead of your real name	MEDIUM
Suspicious URL	HTTP, IP address, or misspelled domain name	HIGH
SPF / DKIM / DMARC fail	Header authentication checks all fail	HIGH
Unusual Reply-To address	Replies go to a completely different domain	HIGH
Grammar or spelling errors	Broken English or awkward phrasing	MEDIUM
Suspicious attachments	.exe, .zip, .docm files from unknown sender	HIGH

6. How to Spot a Phishing Email

Before clicking any link or attachment in an email, ask yourself these five questions:

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| 1. Do I personally know this sender? |
| 2. Was I expecting this email? |
| 3. Does the link URL match the company name exactly? |
| 4. Is this email creating panic, urgency, or a threat? |
| 5. Is it asking for a password, OTP, or card number? |

If you answered NO to questions 1 or 2, or YES to questions 4 or 5:

DO NOT CLICK. Do not reply. Report the email to your IT team immediately.

7. Employee Do's and Don'ts

The following guidelines should be shared with all employees as part of a security awareness program:

DO	DON'T
✓ Check the sender address carefully	✗ Click links in panic or urgency
✓ Hover over links before clicking	✗ Open unexpected or unsolicited attachments
✓ Report suspicious emails to IT	✗ Share OTPs or passwords via email
✓ Use multi-factor authentication (MFA)	✗ Enter your password from an email link
✓ Go directly to the website instead	✗ Trust threats of account closure or fines
✓ Verify requests by calling the company	✗ Assume an email is safe because it looks real

8. Prevention Guidelines

Technical Controls

- Enable SPF, DKIM, and DMARC on your organization's email domain
- Deploy an email security gateway (e.g. Microsoft Defender for Office 365)
- Block executable file attachments at the email server level
- Enable multi-factor authentication (MFA) for all employee accounts
- Configure email clients to show full sender addresses, not just display names

Employee Training

- Run quarterly phishing awareness training sessions
- Conduct simulated phishing tests to measure employee readiness
- Add a one-click 'Report Phishing' button to your email client
- Reward employees who correctly identify and report phishing emails
- Train employees to verify unusual requests by phone — never by email

Incident Response

- Establish a clear, documented process for reporting suspected phishing
- Isolate affected accounts immediately if credentials may have been entered
- Force password resets for any account that clicked a phishing link
- Document and review every phishing incident to improve future defenses

